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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	HouseRent – House Rental Management System
Maximum Marks	5 Marks

Coding & Solution

The HouseRent project is implemented using the **MERN (MongoDB, Express.js, React.js, Node.js)** technology stack. The application provides a secure and user-friendly platform for property owners to manage rental properties and for tenants to search and book available houses. The solution follows a modular architecture, RESTful APIs, and JWT-based authentication to ensure maintainability, scalability, and security.

Instructions:

1. The source code is organized into separate Client and Server modules following the MERN architecture with a clean folder structure.
2. The solution successfully addresses the core problem statement by providing secure authentication, property management, property search, and booking functionalities.
3. The project includes all required source files, configuration files, package files, and setup instructions necessary to install and run the application.
4. The complete source code is maintained in a GitHub repository for version control and collaborative development.

Solution Summary

Field	**Details**
Repository Link / URL	https://github.com/harishvedicharla/house-rent.git
Programming Language(s)	JavaScript, HTML5, CSS3
Framework(s) Used	React.js, Node.js, Express.js, Bootstrap, Vite
Database	MongoDB with Mongoose
Key Features Implemented	• JWT Authentication• User Registration & Login • Role-Based Access Control (Admin, Owner, Tenant) • Property Management (Add, Update, Delete, View) • Property Search & Filtering • Property Booking System • Owner Dashboard • Admin Dashboard
Pending / Incomplete Features	• Online Payment Gateway Integration • Google Maps Integration• Email Notifications • Property Reviews & Ratings • AI-Based Property Recommendation

Setup / Run Instructions	**Step 1:** Clone the repository from GitHub. **Step 2:** Navigate to the **source code** directory. **Step 3:** Install dependencies using **npm install** in both **Client** and **Server** folders. **Step 4:** Configure the MongoDB database connection. **Step 5:** Start the backend using **npm start**. **Step 6:** Start the frontend using **npm run dev**. **Step 7:** Access the application through the browser.

Code Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes/components	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows the **MVC (Model-View-Controller)** architecture.
- JWT and bcrypt.js are used to provide secure authentication and password encryption.
- React components, Express routes, middleware, and MongoDB models are modular and reusable.
- Git and GitHub are used for source code management and version control.
- The project follows MERN Stack development best practices, making it scalable, maintainable, and easy to extend.